

HDOC Day-11 - Menu-Based C Programming

Digit Operations Using Loops and Switch-Case

Batch: B16

Program: B.Tech Computer Science and Engineering

1. Aim

To develop and test a menu-based C program that uses switch-case to perform different digit calculations on a positive integer with while, for, and do-while loops.

2. Problem Statement

- Find the sum of squares of the digits using a while loop. Example: Number 123, Sum = 14.
- Find the sum of cubes of the digits using a for loop. Example: Number 123, Sum = 36.
- Find the difference between the sum of cubes and the sum of squares of the digits using a do-while loop. Example: Number 123, Difference = 22.

3. Algorithm

1. Start.

2. Input a positive integer n.

3. Display a menu containing the three operations and input the user choice.

4. Use switch(choice) to select the required operation.

5. For choice 1, initialize sum to 0. Using a while loop, extract each digit with % 10, add digit * digit to sum, and remove the last digit using / 10. Display the sum.

6. For choice 2, initialize sum to 0. Using a for loop, extract each digit and add digit * digit * digit to sum. Display the sum.

7. For choice 3, initialize squareSum and cubeSum to 0. Using a do-while loop, calculate the square sum and cube sum for all digits. Compute difference = cubeSum - squareSum and display it.

8. For any other choice, display an invalid choice message.

9. Stop.

5. Test Case Table

Test Case	Number	Choice	Operation	Expected Output	Actual Output	Result
1	123	1	Sum of squares	14	14	Pass
2	123	2	Sum of cubes	36	36	Pass
3	123	3	Difference	22	22	Pass
4	45	1	Sum of squares	41	41	Pass
5	45	2	Sum of cubes	189	189	Pass
6	45	3	Difference	148	148	Pass

Verification for number 123: Squares = $1^2 + 2^2 + 3^2 = 14$; Cubes = $1^3 + 2^3 + 3^3 = 36$; Difference = $36 - 14 = 22$.

6. C Program

```
#include <stdio.h>
```

```
int main()
{
    int n, temp, digit, choice;
    int sum, squareSum, cubeSum, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum of squares of digits\n");
    printf("2. Sum of cubes of digits\n");
    printf("3. Difference between sum of cubes and sum of squares\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;
            sum = 0;
            while (temp > 0)
            {
                digit = temp % 10;
                sum = sum + digit * digit;
                temp = temp / 10;
            }
            printf("Sum of squares = %d\n", sum);
            break;
```

```

case 2:
    sum = 0;
    for (temp = n; temp > 0; temp = temp / 10)
    {
        digit = temp % 10;
        sum = sum + digit * digit * digit;
    }
    printf("Sum of cubes = %d\n", sum);
    break;

case 3:
    temp = n;
    squareSum = 0;
    cubeSum = 0;
    do
    {
        digit = temp % 10;
        squareSum = squareSum + digit * digit;
        cubeSum = cubeSum + digit * digit * digit;
        temp = temp / 10;
    } while (temp > 0);
    difference = cubeSum - squareSum;
    printf("Difference = %d\n", difference);
    break;

default:
    printf("Invalid choice.\n");
}
return 0;
}

```

7. Compilation and Execution

Save the source file as q.c, then compile and execute it from the terminal with:

```

gcc q.c -o q
./q

```

8. Output Screenshots

The following screenshots show successful execution for input 123 with each of the three menu choices.

Screenshot 1 - Choice 1: Sum of Squares

```
(kali㉿kali)-[~/Desktop]
$ ./q1
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares = 14
```

Screenshot 2 - Choice 2: Sum of Cubes

```
(kali㉿kali)-[~/Desktop]
$ ./q1
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes = 36
```

Screenshot 3 - Choice 3: Difference between sum of cube and sum of squares

```
(kali㉿kali)-[~/Desktop]
$ ./q1
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference = 22
```

9. Result

The program compiled and ran successfully for all selected operations. The switch-case structure correctly directs execution to the required while, for, or do-while loop, and the observed outputs agree with the test-case calculations.